PlaySBC Service Network Architecture

Readable architecture pack for SIP, RTP, B2BUA, SIPp regression, and RTPengine experiments.

Color key

Config · SIP control · Routing · Internal media · RTPengine · Future · Evidence

PlaySBC

Playful Session Border Controller: break SIP here, not in production.

Professional warning: if a SIP ladder looks like abstract art, the PDF gets redesigned.

Current Lab

UDP/TCP SIP B2BUA, registrar routing, G.711 media, internal transcoding, SIPp regression.

Media Backends

Core profiles use PlaySBC internal RTP. RTPengine profiles anchor media externally.

Evidence

One bundle per testcase: SIP logs, media logs, transcoding logs, PCAP, and HTML report.

Configuration

Helm values render one temporary YAML config per SIPp profile.

ESBC Direction

Static trunks, E.164 policy, registration-backed routing, negative flows, and load tests.

Future Work

TLS, trunk failover, QoS metrics, Kubernetes lab, WebRTC gateway, and AI voice gateway.

How to read this PDF: colored boxes show responsibility. Numbered flows show execution order. Tables explain every logical node.

Zoom note: diagrams are vector-drawn, not screenshots. Use Preview/Chrome/GitHub zoom and the PDF outline for navigation.

Contents And Navigation

Compact diagrams are split across pages so you can zoom into each section without arrow soup.

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Navigation tip: open the PDF outline/sidebar and jump by section. If something looks small, zoom in; the content is vector and should stay sharp.

Enterprise SBC Reference Model

Vendor-style architecture vocabulary, mapped to what PlaySBC implements today and where it is going.

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Design note: inspired by enterprise SBC product docs; this is PlaySBC's own lab model, not a vendor diagram copy.

1 Access Edge
SIPp A, registered caller, future SIP phone/WebRTC client.

→

2 Ingress Security
Digest auth today; TLS, topology hiding, malformed SIP checks later.

→

3 SBC Core
B2BUA dialog state, transactions, routing, registrar, policy.

→

4 Media Services
Internal RTP today; RTPengine anchoring and transcoding experiments.

→

5 Peer Edge
SIPp B, static trunk, E.164 route, future PBX/carrier trunk.

Management Plane
Helm YAML, local runtime config, future Kubernetes Secrets and probes.

Observability Plane
SBC logs, SIP ladder, media logs, PCAP, HTML regression report.

Lab Automation Plane
SIPp regression profiles, RTPengine preflight, pass/fail/blocked verdicts.

SBC-product architecture usually separates control, media, policy, and operations. PlaySBC now mirrors that mental model in lab form.

Product-doc block	PlaySBC mapping today	Future target
Access interface	SIPp UAC/UAS over UDP/TCP	SIP TLS, WebRTC access
Policy/routing	registrar, static route, E.164 profile	trunk groups, normalization, failover
Media service	internal RTP and RTPengine backend	RTCP/QoS, stronger transcoding validation
Operations	logs, PCAP, latest.html	metrics dashboard and trend reports

2. Broad Platform View

A cleaner big-picture view with layers instead of crossed arrows.

Color key

Config | SIP control | Routing | Internal media | RTPengine | Future | Evidence

1 **Access / Test Clients**

SIPp A, registered caller, future SIP phone, future WebRTC browser.

2 **PlaySBC SIP Control Plane**

SIP listener, digest auth, registrar, route policy, dialog state, B2BUA leg manager.

3 **Media Plane**

Internal RTP relay for core profiles, or RTPengine for media anchoring/transcoding experiments.

4 **Peer / Trunk Side**

SIPp B, registered callee, static SIP trunk, E.164 route, future PBX/carrier trunk.

5 **Config / Evidence**

Helm YAML config, logs, PCAP, latest HTML report, future metrics dashboard.

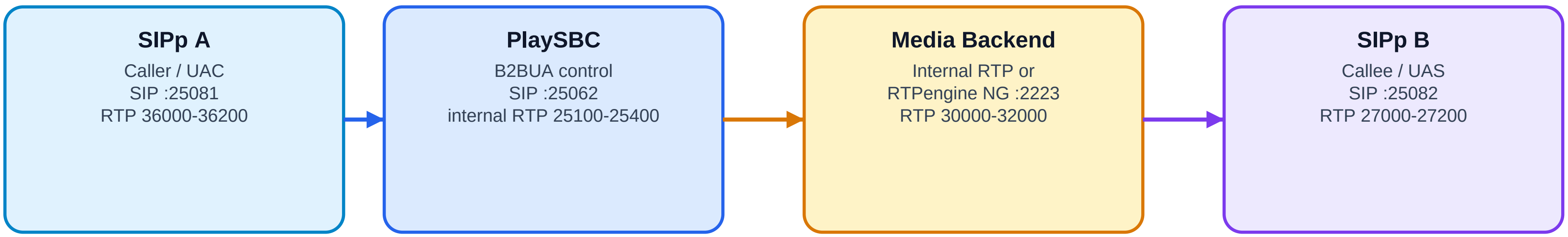
Core idea: PlaySBC owns SIP signalling, routing, dialog state, and evidence. RTPengine is optional and owns media relay/anchoring when selected.

3. Current Local Service Network

Default local ports and responsibilities used by the SIPp regression lab.

Color key

Config SIP control Routing Internal media RTPengine Future Evidence



Config Path

Terminal -> regression runner -> helm template -> server-config.yaml

Control Path

REGISTER / INVITE / ACK / BYE always traverse PlaySBC.

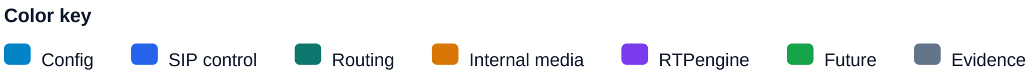
Evidence Path

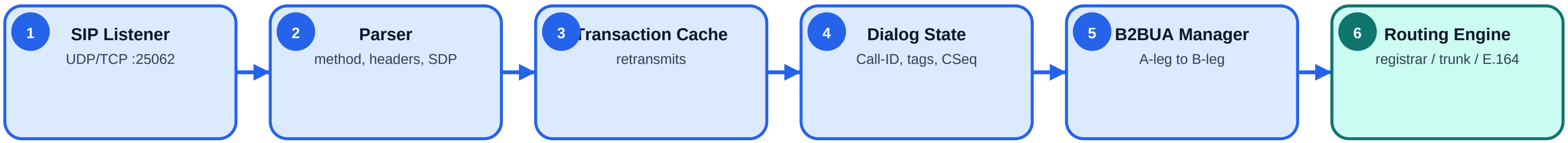
log.sip, log.media, log.transcoding, capture.pcap, latest.html

Important split: SIP signalling always goes through PlaySBC. RTP uses PlaySBC internal media in core profiles and RTPengine in RTPengine profiles.

4. SIP Control Plane Inside PlaySBC

The SIP path is shown as a straight processing chain, with side services explained below.





Side service	Responsibility	Example
Digest Auth	Challenges REGISTER when users are configured.	401 then REGISTER with Authorization.
Registrar	Stores live Contact for registered users.	callee -> sip:callee@127.0.0.1:25082
Route Policies	Select registrar, static trunk, or E.164 route.	+1800* -> sip:{user}@127.0.0.1:25080
Reject Unknown Routes	Makes the lab behave more like an SBC.	Unknown destination returns 404 instead of demo fallback.

5. Media Plane Split

Internal media and RTPengine media are separate choices, not the same path with a different label.

Color key

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Core B2BUA Profiles - Internal Media

Used by basic-media and internal transcoding profiles. PlaySBC receives and relays RTP in the lab. This is useful for learning, packet inspection, and small controlled tests.

RTPengine Profiles - External Media Backend

Used by rtpengine, rtpengine-media, rtpengine-transcoding, and related load profiles. SIP remains in PlaySBC; RTP is anchored by RTPengine.

Aspect	Internal media	RTPengine media
SIP control	PlaySBC	PlaySBC
RTP packets	Through PlaySBC internal RTP relay	Through RTPengine media ports
Transcoding owner	PlaySBC internal logic	RTPengine intent / RTPengine media backend
Expected PlaySBC RTP counter	Non-zero when media flows internally	Zero, because media bypasses PlaySBC

Sarcastic but true: if PlaySBC RTP counters are zero in an RTPengine profile, that is usually good news. Media took the media path, not the scenic route.

6. SIPp Regression Flow

Actual runner behavior: suite-level orchestration calls a per-profile B2BUA runner.

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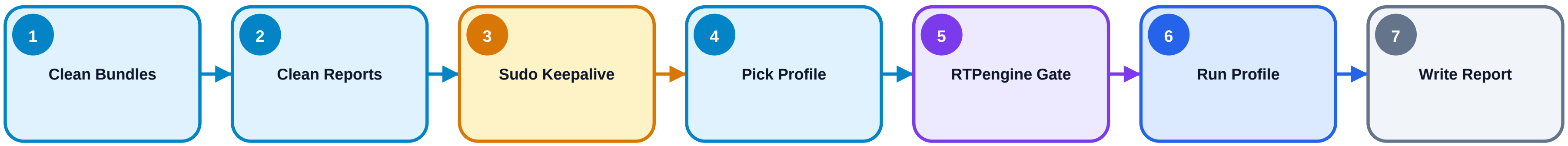
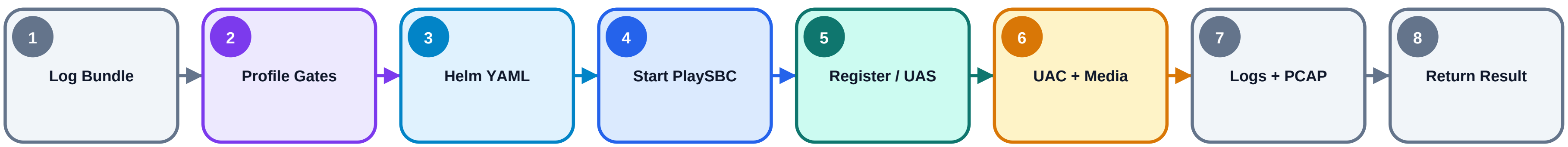
SUITE RUNNER: tools/run_regression_suite.py



PER-PROFILE RUNNER: tools/run_b2bua_sipp_smoke.py



Layer	What actually happens
Suite runner	Cleans old passed/blocked B2BUA bundles, cleans old reports, starts sudo keepalive when PCAP replay needs it, loops selected profiles, gates RTPengine profiles, runs the profile command, then writes latest.html.
Profile runner	Creates one log bundle, performs profile-specific gates, renders Helm values into server-config.yaml, starts PlaySBC, registers endpoints, starts SIPp B, runs SIPp A and media, collects logs/PCAP/results, then returns one profile verdict.
Important	Helm config rendering happens inside the per-profile runner before PlaySBC starts. RTPengine can be gated at suite level and again inside the profile runner.

7. Basic B2BUA Call Flow

A compact ladder table is easier to read than diagonal arrows over the content.

Step	SIPp A	PlaySBC B2BUA	SIPp B
01	INVITE ->	receive INVITE; send 100 Trying	
02		create outbound INVITE ->	receive INVITE
03		<- 100 Trying / 180 Ringing / 200 OK	send provisional/final response
04	<- 180 Ringing / 200 OK	relay response to A-leg	
05	ACK ->	receive ACK; forward ACK ->	receive ACK
06	RTP	internal media or RTPengine anchored path	RTP
07	BYE ->	200 OK to A; send BYE ->	receive BYE
08		<- 200 OK	send 200 OK

Read it like a real SBC: PlaySBC terminates the A-leg and originates the B-leg. It is not merely a UDP pipe with confidence.

8. Logical Node Catalog - Lab And Service Nodes

Every logical node used in the diagrams, with its technical meaning and evidence.

Node	Meaning	Example	Evidence
Developer Terminal	Human starts checks, RTPengine, sudo cache, and regression.	Full local command from README.	Terminal output
Regression Runner	Sequentially executes SIPp profiles.	tools/run_regression_suite.py	latest.html
Helm Template	Renders YAML config for local profile runs.	helm template charts/playsbc	server-config.yaml
Per-profile Config	One runtime YAML per profile.	media_backend: rtpengine	log.platform
SIPp A	Caller side / UAC / registered caller.	INVITE to :25062	log.sipp, capture.pcap
SIPp B	Callee side / UAS / registered endpoint.	100/180/200 OK	log.sipp, capture.pcap
RTPengine	Optional external media anchor/backend.	NG UDP :2223	log.media, query stats
HTML Report	One row per testcase/profile.	logs/reports/latest.html	PASS/FAIL/BLOCKED

9. Logical Node Catalog - PlaySBC Internals

The main internal components and what each is responsible for.

Internal node	Responsibility	Example	Evidence
SIP Listener	UDP/TCP socket entry point.	REGISTER, OPTIONS, INVITE, ACK, CANCEL, BYE	log.sip
SIP Parser	Extracts headers, body, Call-ID, CSeq, tags, Contact, SDP.	m=audio from SDP	log.sip
Transaction Cache	Handles retransmitted requests consistently.	duplicate INVITE branch/CSeq	retransmission profile
Dialog State	Tracks A-leg and B-leg identity.	Call-ID, From tag, To tag	log.call
Digest Auth	Challenges REGISTER when users exist.	401 + Authorization	registration ladder
Registrar	Stores live Contact routes.	callee -> sip:callee@127.0.0.1:25082	registered profiles
Routing Engine	Chooses registrar/static trunk/E.164/reject.	+1800* policy	log.platform
B2BUA Manager	Creates outbound B-leg and relays responses.	A-leg INVITE -> B-leg INVITE	log.sip ladder
Internal RTP Relay	Lab RTP path for core profiles.	G.711u/G.711a	log.media
RTPengine Client	Controls external media backend.	offer, answer, query, delete	log.media

10. Config And Evidence Map

Where each important behavior is configured and where to prove it worked.

Core B2BUA config

```
sip_transport: udp
media_backend: internal
default_codec: PCMU
route_policies:
  - name: registered-endpoints
    match: "*"
    target: registration
```

RTPengine config

```
media_backend: rtpengine
rtpengine_url: udp://127.0.0.1:2223
rtpengine_timeout: 3.0
b2bua_ladder_logs: true
reject_unknown_routes: false
```

Evidence file	What it proves
log.sip	SIP messages, ladder, A-leg/B-leg view
log.media	RTP observations, RTPengine offer/answer/query
log.transcoding	Codec direction and transcoding owner
capture.pcap	Single-call SIP/RTP packet evidence
latest.html	Regression result summary

Tiny bit of attitude, serious rule: YAML without logs is just a wish. PlaySBC tries to leave receipts.

11. Future Enhancement View

A broader architecture direction, still anchored in testability and evidence.

1 **SIP Transport Hardening**
TLS, TCP connection reuse, transport-specific policies.
Status: Near

2 **ESBC Lab Features**
Trunk groups, failover, header normalization, CAC.
Status: Near

3 **Media Quality**
RTCP, jitter/loss, RTPengine health, QoS reporting.
Status: Near

4 **Kubernetes Lab**
Docker image, Helm install, probes, Secrets.
Status: Near

5 **WebRTC Gateway**
SIP WebSocket, ICE/STUN, DTLS-SRTP.
Status: Future

6 **AI Voice Gateway**
RTP -> STT -> LLM -> TTS -> RTP.
Status: Future

Roadmap rule: if it cannot be configured, tested, logged, and explained, it is just a feature-shaped rumor.